GRISHMA BALGI

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Portfolio Link

Education

New York University (NYU) **Aug. 2025 – May 2027**
M.Sc. in Mechatronics and Robotics New York, United States
Coursework: Foundation of Robotics, Mathematics of Robotics, Mechatronics **CGPA: 3.8/4.0**

Veermata Jijabai Technological Institute (VJTI) **Aug. 2021 – Jun. 2025**
B.Tech. in Mechanical Engineering, Honors in Automobile Engineering **CGPA:9.04/10**
Coursework: Robotics, Data Science and Data Analytics, Computer Programming

Skills

CAD: Solidworks, Creo Parametric, CATIA, AutoCAD, Altair Twin Activate, Fusion 360
FEA: Ansys Workbench (Rigid Body Analysis & Static Structural), Altair (Motion View, romAI), Mechanical APDL
Programming: MATLAB, Python (Numpy, Pandas, matplotlib, BeautifulSoup, PyTorch, TensorFlow), C++, C, Tableau
Tools: Flutterflow, Microsoft Office (Excel, Word, Powerpoint), SQL, Google Analytics, Canva, Microsoft
Robotics: Robot Operating System (ROS/ROS2), Gazebo, MuJoCo, RoboAnalyzer, Creality, Cura Ultimaker
Hardware Arduino Mega/Uno, Raspberry Pi, Sensors, Microcontrollers, PCB Design

Experience

Schneider Electric **Jun. 2024 – Jun. 2025**
Machine Learning Research Intern Mumbai, India
- Developed digital twins and hardware prototypes of low voltage circuit breakers to enable production readiness and predictive maintenance. Collaborated with cross-disciplinary teams to integrate sensors and troubleshoot embedded electronics for live-testing environments.
- Designed a real-time fault detection system for circuit breakers, integrating sensors and automation hardware to support continuous improvement of equipment reliability; achieved 99% classification accuracy using machine learning models.
- Developed and deployed Machine Learning models using Pytorch and TensorFlow to embedded systems, optimizing performance and achieving 99% improvement in processing speed.

VJTI Racing **Apr. 2022 – Apr. 2024**
Vice-Captain & Joint General Secretary Mumbai, India
- Led cross-functional team in the design, assembly, and integration of subsystems for an electric all-terrain vehicle. Led design, mechanical assembly, wiring, and testing of automated subsystems for an electric all-terrain vehicle. Strengthened my skills on Solidworks, Ansys, AutoCAD, 3d printing (FDM) and Creo Parametric.
- Secured sponsorships worth INR 20 lakhs and managed the complete annual budget, vendor negotiations, procurement, and reporting.

Projects and Publications

Gesture Controlled Robotic Arm: Media Pipe, Arduino IDE, Python
- Designed and deployed a closed-loop PD control system for a gesture-driven robotic finger, improving actuation response time by 50% and reducing settling time from 0.25 s to 0.12 s.
- Enhanced tracking accuracy through custom servo feedback integration and digital controller discretization (Tustin) on an Arduino-based embedded platform.

Multi-Sensor Setup for Fault Detection of Circuit Breaker: Python, Arduino IDE
- Designed a real-time fault detection system using Arduino Nano 33 IoT, vibration and sound sensors, and python-based data logging; achieved 99% classification accuracy using machine learning models.
- Designed a digital twin interface for live monitoring and fault classification, enhancing preventive diagnostics and reducing downtime in electrical systems.

Wall-E: Python, Arduino IDE
- Worked on creating a line-following and self-balancing bot using ESP32 module. Further studied concepts on Line Sensor Array, Pulse Width Modulation Signals and PID controller.
- Analysed flowcharts on self-balancing robots algorithms and also generated an OLED display integrated with the bot.

Obstacle Detection and Automated Steering using Python and MATLAB Simulink: Python, MATLAB Simulink
- Designed a multi-lane autonomous driving simulator combining virtual LiDAR sensing and machine learning for real-time obstacle avoidance. Trained a decision tree classifier on 168,000+ samples, achieving 99.94% accuracy in predicting safe lane change actions.
- Deployed the model with a MATLAB-based user interface featuring dynamic lane shifts, collision alerts, and live trajectory visualization.